Film Studio
UX & Interface

Desktop three-panel production. Voice-led briefing. One multimodal composer for uploads, slash skills and contextual references.

UX architecture | Screen designs | Component contracts | Acceptance tests

For Codex implementation review
Design proposal · no deployment or paid operations authorized

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Contents

Full-size visual plates follow the detailed specification. The source Markdown and standalone HTML prototype accompany this PDF.

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Director Studio | Customer-Facing UI/UX Specification v2.0

Status: Design proposal and implementation handoff; not an authorization to deploy or spend.
Surfaces: Desktop web, responsive tablet, mobile web.
Primary launch track: Short films; shared foundation anticipates music videos and advertising.
Companion: `FILM_STUDIO_BLUEPRINT.md`. This is the **customer-facing** product, not the administrator backoffice.
Artifacts: Working demonstration HTML and five rendered reference images accompany this specification. All sample films, uploads, chats and UI actions are illustrative.
Design decision: Reusable multimodal composer; desktop project/canvas/Director workspace; voice or text; skill discovery through `/`; project references through `@`; uploaded assets through `+`; explicit approvals and spending authorization.

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1. Executive decision and design boundaries

Director Studio is a film-creation environment, **not a prompt box wrapped around model APIs**. A newcomer should be able to explain an idea, attach examples, reach a reviewable treatment and storyboard, and understand the next consequential decision. Experienced users should be able to bring scripts, images, audio and existing footage without repeating the first-time interview.

The authoritative flow remains: **Brief -> Story -> Cast & World -> Storyboard / Animatic -> Production -> Edit & Review -> Export.** Each stage has versioned artifacts, acceptance criteria and the possibility of returning to an earlier stage without silently invalidating accepted work. Other production formats have distinct intake/timing logic but reuse the project, asset, approval and job framework.

Non-negotiable rules:

- Ordinary creative information and the next decision are visible. Background orchestration and technical activity are **collapsed by default**; `View Activity` exposes an understandable event summary only when clicked. Never reveal raw hidden model reasoning or privileged implementation logs.
- The brand or name of any external orchestration framework must never appear in the product interface, user help, product documentation, or management panels. Implement desired patterns natively.
- Adding references, selecting `/skills`, mentioning `@assets`, editing a brief or recording speech **never launches paid generation implicitly**.
- Show known/estimated additional charges before a paid operation, and require authorization under the current project budget. Unknown price blocks submission until it can be estimated/authorized.
- Protect approved constraints, shots and neighboring assets. A correction targets affected versions rather than rerendering an entire film.
- Do not represent model features as guaranteed: character-lock, camera precision, audio synchronization and automatic defect repair all require endpoint- and output-specific verification.
- The demo is not connected to real voice services, private storage, a production database, payment authorization, or live model generation.

1.1 Scope of this document

This document expands the previously discussed UI Factory direction into layout rules, page flows, micro-interactions, screen states, accessibility requirements, data contracts and developer acceptance tests. It does not replace the film-production architecture. Separate technical decisions about the job queue, storage, provider schema and rendering remain governed by the companion blueprint.

1.2 Priority classes

Priority	Meaning	Representative features
P0	Necessary to validate the customer film workflow	Uploads with roles, composer, `/` picker, `@` references, typed messages, shot context, auth, project persistence, shot approvals, budget gates, responsive review
P1	Important for differentiated creation and early retention	Tap-to-talk transcription, spoken replies, targeted shot repair, comparison, richer reference inspection, full keyboard/clipboard media handling
P2	Only after core usability/economics are proven	Opt-in continuous realtime Director, live multiuser co-creation, advanced node canvas, granular mobile multitrack editing, expressive talking avatar

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2. UX principles and research interpretation

Current public product patterns from Runway, Krea and Hedra show that people expect conversation, references, assets, generated media and editing to stay near each other. Published product materials describe slash/mention interactions in some modes, reusable references and accessible creation panels; **do not assume their implementations, entitlements or exact three-way geometry are identical**. Our response is task-driven: three adjustable panels on desktop, two on medium screens, one focused canvas on phones.

UI Factory framework: task-first interaction hierarchy with Material 3 conventions for controls, modal/sheet behavior, keyboard focus, responsive reflow and status indicators. An editorial light surface suits first-use briefing. Graphite media surfaces suit detailed editing; restrained blue indicates selected actions. No fluorescent neon, fake futurism, oversized decorative hero, ornate typography or content-free cards.

Design-test hypotheses, not proven conversion results: the first film treatment should arrive without compulsory model selection; optional voice lowers input friction; scoped references reduce unexpected character drift; preview-before-render reduces avoidable spending. Validate these hypotheses with real users and production-cost data.

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3. Users, journeys and FTUE

3.1 Two primary profiles

First-time creator: knows the outcome, not filmmaking vocabulary. Enters through homepage, chooses Short Film, speaks or types a premise, optionally attaches references, answers only consequential questions, reviews a treatment and a timed visual plan, then decides whether to spend on production.

Returning/professional creator: opens Recent Projects or `Create from assets`, uploads an existing script/shot list/song/reference pack, inspects extracted content, corrects misclassified asset roles, optionally opens advanced shot specifications, and returns to the current edit or approval with state preserved.

3.2 Customer FTUE flow

Step	Primary question	User sees	Primary action	Recovery
1. Welcome	What should I make?	Short Film, Music Video, Advertisement; voice/text entry	Describe film	Change format without losing text
2. Brief	What story do I mean?	Director conversation and editable extracted constraints	Continue to treatment	Correct transcription/ reference roles
3. Story	Does the story work?	Treatment, scenes, dialogue and story change	Approve story	Request rewrite without video spend
4. Cast & World	Who/where must remain stable?	Approved reference candidates and intended use	Approve identity/ world	Replace or exclude source image
5. Storyboard	Will the film read?	Shot images, specifications and animatic timing	Approve and get estimate	Revise individual shot/sequence
6. Production	Are outputs acceptable?	Generated takes and review findings	Accept or request targeted revision	Reconcile failures; preserve accepted assets
7. Edit & Export	Is the whole film ready?	Rough/final cut and delivery options	Approve and export	Fix sound/captions or retry export only

Account gate: let visitors begin the brief, but require sign-in before persisting private assets, inviting collaborators or authorizing paid work. If sign-in interrupts a drafted prompt, preserve it after successful authentication where technically supported. Do not assume anonymous uploads are permanently stored.

FTUE success metric (proposed): a newly signed-in user reaches a **reviewable, editable story treatment and storyboard** with understood constraints. A first playable export is a later, separately reported activation milestone. Do not invent a time-to-first-film commitment.

4. Information architecture and navigation

4.1 Top-level customer destinations

- **Home:** new project, resume, recent projects. This is not the Admin Today dashboard.
- **Projects:** searchable project cards with status, last edited, accepted assets and next required action.
- **Library:** user/workspace-permitted reusable characters, locations, scripts, music and uploaded media, with licenses and reference roles.
- **Workspace:** selected project and stage. Primary tabs: Story, Cast & World, Storyboard, Production, Edit, Review & Export.
- **Account / Usage:** plan/entitlement, project spend, security, microphone preferences and storage policy.

4.2 Screen inventory

Screen	Desktop treatment	Mobile treatment	P
Public homepage	Light editorial entry + Short Film featured	Narrow text/voice entry + format selector	P0
Signed-in home	Recent projects, create action, continue work	Project cards, prominent voice entry	P0
Voice/text briefing	Conversation plus evolving brief/reference panel	Conversation and dedicated composer sheet	P0/P1 voice
Story editor	Structured treatment and scene list	Readable story cards and change request	P0
Cast & World	Reference gallery and role/ consent inspector	Swipeable character/world cards	P0
Storyboard	Shot grid, detail inspector and animatic	Shot-by-shot review with clear approval	P0
Production studio	Three-pane project/canvas/ Director	Canvas + Director bottom sheet + reference drawer	P0
Editing	Centre timeline, media viewer and sound controls	Simple trim/annotate/ select; desktop for complex edits	P1
Final review/export	Playable cut, comments, output parameters	Playback, review and export action	P0
Media library	Search, filter, reference role, lineage	Filter and select existing asset	P0
Usage/settings	Itemized estimates/ actuals, account	Focused usage and settings pages	P0

5. Visual design system

5.1 Surfaces and typographic rules

Token	Suggested value	Use
`--canvas-light`	`#FAFAF8`	Briefing/home base
`--light-panel`	`#FFFFFF`	Cards and attachment previews
`--light-border`	`#E0E4E8`	Subtle boundaries
`--studio-bg`	`#191C21`	Media canvas surround
`--studio-panel`	`#22262D`	Project and Director panels
`--studio-border`	`#3C424C`	Studio separators
`--primary`	`#355D9D`	Selection, authorizing navigation
`--text-dark`	`#242B33`	Light body text
`--text-light`	`#EDF0F3`	Dark body text
`--positive`	`#9BC1AB`	Reviewed/succeeded state, with text label

Typography: a standard reliable sans for functional UI (proposed **IBM Plex Sans** where permitted, with system sans fallback), plus an unobtrusive serif heading only in the editorial briefing; avoid decorative handwriting or trendy AI landing-page conventions. Ensure desktop primary body text normally 13–15 CSS px, mobile main text at least 14px and readable labels, subject to verified responsive/accessibility review. Do not use a colored badge as the only indicator of approval.

5.2 Spacing and constraints

- Base spacing unit: 4 CSS px; controls typically 36–44px high on desktop and touch controls typically at least 44px on mobile.
- Corner radii: 8px inputs/controls, 10–12px panels, 16px modal and bottom sheet.
- Borders and modest elevation separate functionally distinct zones; no gratuitous shadows.
- Use aspect-ratio-preserving media with letterboxing rather than cropping approved shots to fill a viewport.
- Icons always have accessible names. Do not use hover to reveal an essential function on touch devices.

6. Desktop Production Studio: three-pane layout

> **PLATE A** Desktop Production Studio: three-panel workspace. See full-resolution visual plates at the end of this PDF and the separate PNG files in the implementation package.

Entry: existing project in Production, with selected scene/shot and at least one take.
Success: inspect the take, add references or a skill, ask for a constrained correction, review cost and accept/deny without altering unrelated accepted footage.

6.1 Grid geometry

At a 1440px viewport use approximately **18% left / 57% canvas / 25% right** (excluding global header); target 20/55/25 is an acceptable starting alternative. The centre must have a practical minimum of approximately 560px for detailed editing, otherwise collapse optional side content. Proposed min/max constraints: left 220–340px, right 300–480px, centre flexible. Persist individual user panel preferences per device class if authorized; do not force a phone to inherit desktop widths.

Resize behavior: dragging a splitter should update local widths, preserve focus/selection and avoid resizing below useful media controls. Double-click or `Reset layout` restores defaults. Keyboard resize commands/accessible handles are required if draggable controls ship; otherwise use collapse/expand controls first.

6.2 Left pane: project intelligence, not a generic file explorer

Contents in descending priority: project identity and status, numbered scenes, selected shot/sequence, Cast & World thumbnail grid, Project References, optionally asset library and current production-state summary. Clicking a character or location opens its reference inspector; it must not silently attach an unrelated asset to the next generation.

Reference inspector: asset preview, filename/type, declared role, project/scene/shot scope, version, approved/rejected state, source/licensing/consent status, and `Use in this shot` action when allowed. Deleting a reference is distinct from removing its use in a single message.

Use collapse groups for long productions. Search/filter scene lists with stable scene IDs, not brittle display positions.

6.3 Centre pane: media-first and stable

Toolbar shows project > scene > shot breadcrumbs, selected take, available compare mode and review status. Main area uses a video/image player with meaningful play/seek, comparison and frame annotation affordances. Below it show the selected scene's timeline/shot strip, with dialogue/score/ambience rows only when present. An approved take may still require sound or edit approval; represent these independently.

When there is no generated take: show the approved storyboard frame, requirements and **estimated** next-generation cost if verified. Never show a fabricated completed video frame or progress percentage.

Actions: `Compare takes`, `Request change`, `Approve take`, `View shot specification`, `Review adjacent shot` and contextual export actions. Each action is active only when its prerequisites are satisfied.

6.4 Right pane: AI Director and composer

Top: Director label, currently scoped item, optional `View Activity`. Middle: concise conversation and current-shot decision cards. Bottom: sticky multimodal composer. Do **not** permanently expose agent-to-agent logs. The right pane can collapse for media review; a compact `Talk to Director` launcher reopens it with the exact selected shot.

Selection context must be explicit, e.g. `Scene 03 / Shot 06 / Take 02`, with a clear option to change scope to scene/project before submitting. The Director must not infer arbitrary scope from vague chat history.

6.5 Primary desktop empty/failure cases

- **No project:** show `Create a film` and `Open project`, not empty production metrics.
- **No selected shot:** select from the scene list or offer guided storyboard creation.
- **No generated take:** show approved board, required reference checklist, and `Prepare generation` only after price check.
- **Failed/unknown provider submission:** display recorded job status and non-billable `Check status`; do not offer blind repeat generation.
- **Reference missing/expired:** explain how to restore it and which downstream work is affected.
- **Budget exhausted:** suggest revised estimate/new authorization, never auto-charge.

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7. Desktop Voice/Text Briefing

> **PLATE B** Desktop Voice Briefing: conversation and film brief. See full-resolution visual plates at the end of this PDF and the separate PNG files in the implementation package.

The briefing layout deliberately departs from the production layout. A persistent stage list (left), conversational briefing (centre) and evolving film brief/reference inspector (right) are more useful here than a large empty video player.

First-time state: one welcoming question, example creative formats, voice/text entry and attach affordance. Avoid more than one or two clarifying questions at a time. Critical details: format, premise, intended runtime, language, characters/location if relevant, and desired emotion/style. Do not insist on advanced model names, frame rates or lenses before the idea exists.

Returning state: open the existing script/treatment, show what was extracted, ask permission before replacing user-authored lines, and explain any incompatible new reference. Reference chips remain attached to the draft and also appear in the right-hand `Film brief` summary.

Controls: `+ Attach`, `/ Skills`, `@ References`, microphone, Send, Continue to Story. The main Send action drafts/proposes a plan; it does not launch a paid motion job.

Voice design: tap-to-talk is recommended for first release, using visible states `Idle`, `Requesting microphone permission`, `Listening`, `Transcribing`, `Transcript ready`, `Sending`, `Speaking response`, `Error`. Display editable transcript before send for consequential instructions. Spoken replies are optional and off by default in public/shared settings. Long continuous real-time sessions are opt-in, cost-managed and deferred until proven valuable.

Story treatment gate: the user can correct the draft, review generated scenes, and accept a **specific artifact version**. An unaccepted plan is not permitted to trigger paid generation.

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8. Mobile: focused canvas and Director bottom sheet

> **PLATE C** Mobile: selected-shot Director composer. See full-resolution visual plates at the end of this PDF and the separate PNG files in the implementation package.

Recommended mobile default: full-width selected-shot preview, short review summary and a persistent `Talk to Director` action that opens a context-aware bottom sheet. Three permanent desktop columns are forbidden on small screens. Desktop-style advanced editing remains optional, not crammed into 390px.

Mobile functions: choose previous/next shot, approve/reject selected take, open references, attach from device/photo library, use Skills button or type `/`, mention references with `@`, tap microphone, review transcript, inspect budget impact and authorize a targeted correction. The bottom sheet must avoid covering the keyboard, safe-area indicator and its own controls; expand and scroll within available viewport height.

Mobile skill menu state:

> **PLATE D** Mobile: slash-skill picker. See full-resolution visual plates at the end of this PDF and the separate PNG files in the implementation package.

Mobile attachment menu state:

> **PLATE E** Mobile: attachment picker. See full-resolution visual plates at the end of this PDF and the separate PNG files in the implementation package.

Tablet recommendation: two working columns at sufficient width (canvas and Director) with project assets as a drawer. At approximately 1000px and below, prioritize media width over retaining all three panes. Exact breakpoints should be tested based on actual content, not device names alone.

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9. Universal multimodal composer: component specification

9.1 Component scope

One `DirectorComposer` should be reused by Homepage/FTUE, Briefing, Storyboard, Production and Edit. Props provide `workspaceId`, `projectId`, optional `sceneId`, `shotId`, `takeId`, `stage`, `mode`, `allowedSkills`, `acceptedAssetRoles`, `permissions`, `budgetPolicy`, and a `draftId`. It should not contain business logic for charging providers.

Visible anatomy:

1. Input placeholder that adapts to context: `Describe your film` on FTUE versus `Ask Director about Shot 06` in Production.
2. Token row for referenced project assets, attachments and selected skills. Tokens show filename/short label and declared role.
3. Toolbar: `+` attachment, `/` skill, `@` existing references, microphone, send. Label controls on mobile or provide accessible names and discoverability.
4. Context indicator showing project/scene/shot; click to inspect or change scope.
5. Draft save state and any upload/transcription/validation error, visible near the control affected.
6. Price/permission warning only when a planned action is consequential; do not decorate every idle composer with billing notices.

9.2 Input types and file pathways

Input	Provisional supported formats	Example role
Still image	JPEG, PNG, WEBP	Character Identity, Wardrobe, Location, Visual Style, Prop, Shot Reference
Video	MP4, MOV, WEBM (verify backend decode)	Action Reference, Camera Motion, Existing Footage
Audio	WAV, MP3, M4A (verify metadata and vendor route)	Master Song, Dialogue Reference, Voice Reference, Ambience
Document	PDF, DOCX, TXT, MD	Screenplay, Treatment, Shot List, Brief
Project library	Authorized records only	Existing character, voice, location, shot, timeline asset

Policy: configure size/length, file count, MIME allowlist and quotas server-side after provider/storage validation. Do not invent universal limits or advertise an unsupported format. Validate MIME using server inspection rather than filename alone. Uploaded documents/images are untrusted input; content must never override system permissions or approve spending.

9.3 Upload sequence and scope

1. User selects/drop-pastes files or chooses Library. Client displays queued file chips immediately.
2. Server validates session, workspace access, provisional MIME/size and quota; returns signed **scoped** upload target if permitted.
3. Client uploads directly to private object storage with progress and cancel (when supported).
4. Server finalizes asset: checksum, MIME verification, metadata, provenance, malware/unsafe-file handling where relevant, license/consent record and durable asset ID.
5. User chooses reference role and scope (project, scene, shot, conversation only). Inference may suggest a role, but must be editable.
6. Composer adds an **asset ID**, not an expiring URL, to the message; generation routing verifies compatibility and access before use.

Upload states: selected, validating, uploading, processing, ready, failed, removed. A failed upload must not destroy message draft; safe retry should reuse the same local selected file where available. Removing a token from a message must not delete a shared library asset unless a separate authorized delete action occurs.

9.4 Reference role and lock semantics

Roles are semantic constraints, not labels only. `Character Identity` points to approved identity references, not a universal guarantee of likeness. `Wardrobe` controls outfit/look. `Location` describes spatial environment. `Visual Style` can be flexible unless formally approved. `Master Song` anchors music-video timeline and must not be tempo-shifted casually. `Shot Reference` governs composition and performance intent where supported.

Store `required`, `preferred` or `illustrative` binding, selected scope and version. The router must report unsupported bindings. Never say an unverified model will preserve exact multi-view identity simply because a reference was attached.

9.5 Message model and activity behavior

Composer messages may combine text, ordered attachment references, asset mentions, a structured selected skill and explicit target context. Store exactly what the user submitted and return a message ID. Distinguish user-authored text from assistant summaries. Background planning activities remain private/collapsed; `View Activity` shows human-readable events and statuses when selected.

When scope changes, preserve unsent text and file chips if allowed, but require explicit review before a request is sent to a different selected shot than originally intended.

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10. Slash-skill picker: exact interaction contract

Trigger: `/` typed at the start of text or at an allowed word boundary opens the picker without submitting. Subsequent characters filter using name, aliases and short description. Do not trigger in an ordinary web URL, pasted code block, or during an active IME composition sequence. The visible Skills button provides the same picker for mobile/touch users.

Pick behavior: search results grouped by `Recommended for this stage` and `Other available` only when useful; each row shows canonical slash command, plain-language purpose, required input types and unavailable reason if disabled. Selecting a row inserts a **structured skill token** (ID/version) into the draft, replacing the typed slash command. A chip can be removed. A skill selection itself costs nothing.

Keyboard: Arrow Up/Down changes active option, Enter selects, Escape closes and returns focus, Tab has documented accessible behavior. Screen readers receive menu state and option descriptions. Empty results offer `No matching skills`; no automatic default selection. A trailing `/` after selecting an option remains ordinary text unless typed anew.

Registry: skills are not hardcoded into the application as prompts. Expose only approved/active versions through a capability endpoint, scoped by project type, stage, role and permission. Example launch candidates: `/storyboard`, `/character`, `/camera`, `/lighting`, `/continuity`, `/lip-sync`, `/fix-shot`. Their names are placeholders pending actual registry and workflow adapter verification. Version pinning is necessary for reproducibility.

Example: user types `/fix-shot Make her expression more restrained; keep the approved camera framing.` The app stores a `fix-shot` skill selection, text, selected take ID, required locked constraints and asset

references. The assistant may propose a repair plan; actual generation requires confirmation and estimate.

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11. `@` project references

Typing `@` at a text boundary, or clicking the mention button, opens a contextual list: Character, Location, Script, Audio, Shot, Previous Take. Only current-workspace-authorized objects appear. A mention becomes a typed token linked to an immutable asset/version ID, not a free-text asset name. A deleted/expired/out-of-scope asset should be shown as unavailable with an explanation.

Important distinction: `+` uploads or chooses an item; `@` mentions/links an existing item in a message; `/` selects a skill. These three actions are deliberately separate even though they share the same composer toolbar.

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12. Conversational voice UX and cost controls

Primary model: tap-to-talk voice recording/transcription, then the same text-and-tools path as typed messages. Short spoken replies are optional. Preserve language, punctuation and proper names for review; support Indian languages as provider capability allows, without claiming identical quality across languages.

Voice state progression:

`idle -> permission_requested -> listening -> stopped -> transcribing -> transcript_review -> submitted -> optional_speaking -> idle`.

Failures branch to recoverable states: permission_denied, device_missing, no_audio, network_failure, transcript_failed, session_expired. Provide type-instead fallback. The microphone button must reflect whether audio is actually recording and must never masquerade as live when only a local demo is available.

Costs and privacy: set reasonable length caps at service-policy level after measurement; do not leave continuous realtime sessions idle during rendering. The client must show an intelligible opt-in for mic access and indication while recording. Voice recordings, transcripts and generated responses follow a published workspace retention and deletion policy; do not reuse customer material for training without specific permissions.

Action safety: voice requests can propose script or shot changes; they cannot bypass approvals, identity/rights checks or budget authorization. Before a paid request, present a concise planned change and require a deliberate action. Critical spoken or recognized numbers should be visible in text before execution.

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13. Shot-level actions, approvals and pricing

Intent -> proposal -> cost/impact preview -> authorization -> job -> retrieval -> review -> accept.

When the user asks to fix a shot, the assistant should produce: selected shot/take, defect diagnosis, mandatory preserved constraints, allowed modifications, compatibility warnings, proposed tool route if verified, estimate/range with assumptions, and list of downstream approvals affected. The `Generate correction` control remains disabled until required approvals and spend authorization are recorded.

Approval matrix: treatment, character identity, wardrobe/look, storyboard frame, shot motion, dialogue/audio, selected take, edited sequence and export are distinct approvals. A storyboard-frame approval does not automatically approve generated motion. Changing a character version flags affected shots, but accepted media is not silently deleted or regenerated.

Budget rule: the backend makes an atomic reservation before any potentially billable submission. Keep unknown provider outcomes reserved until reconciled. A soft refresh or repeated Send action must not create duplicate paid requests. A user can ask `Why was I charged?` through an authorized, itemized view.

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14. Responsive behavior and breakpoints

Available viewport	Proposed layout	Interaction detail
>=1280px	Three active panes	Resizable project / canvas / Director; centre remains dominant
1000–1279px	Canvas + Director, project drawer	Collapse left; preserve selected asset chips
600–999px	Media canvas with collapsible side drawer or two-pane tablet	Avoid fixed 25% pane that becomes unreadable
<600px	Single canvas, Director bottom sheet, context/reference drawer	Touch-first Skills, `+`, microphone and send; no mini desktop timeline

Breakpoints are design starting points, not accessibility proof. Honor safe-area insets, keyboard open/close, focus placement, large text, portrait/landscape and reduced motion. Test minimum supported viewport widths with real content and multiple attachment chips. The selected-shot context must remain visible as a short breadcrumb when the bottom sheet opens.

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15. Screen states and error-copy specification

Event/state	User-facing message	Required action	Never do
First empty project	`Start with a story, a script or a song.`	Type, speak or attach	Auto-create a paid job
Upload rejected by server	`This file cannot be used. Check its type or size.`	Replace/retry without losing message	Silently ignore file

Event/state	User-facing message	Required action	Never do
Upload temporarily failed	`Upload interrupted. Your draft is still here.`	Retry	Lose user text
Permission denied for mic	`Microphone access is off. You can type instead.`	Type or device permission guidance	Pretend recording started
No matching slash skills	`No approved skills match your search.`	Clear filter	Substitute unrelated skill
Skill unavailable for stage	`This skill is not available at this stage.`	Switch stage or inspect reason	Execute unsupported route
Expired reference	`Reference unavailable. Reconnect or remove it.`	Choose alternative	Send expired URL to provider
Shot not approved	`Approve the required storyboard and references first.`	Open prerequisite	Start generation
Price unknown	`We cannot confirm this generation cost yet.`	Wait/choose supported route	Estimate a made-up cost
Provider submission uncertain	`Checking the original request before trying again.`	Check status	Double-submit paid work
Output retrieved	`Take ready for review.`	Play and accept/change	Mark film delivered
Export playback failed	`Export needs repair; approved shots are safe.`	Retry finishing task	Rerender all shots

16. Accessibility and performance

Use WCAG 2.2 AA as the design target, subject to measured implementation validation. Keyboard accessibility must cover sidebar navigation, tab order, `/` skill menu, `@` reference menu, upload controls, modal cost confirmation and closing bottom sheets. Focus remains visible on both themes; status does not rely on color alone. Screen reader labels identify current selection and approval state.

Avoid embedding base64 media in the production HTML; the provided demonstration does so solely to make the visual handoff portable. Production should load thumbnail/previews from authorized asset storage with correct caching, image sizing and lazy-loading. Media preview should not block basic navigation. A long-running provider request cannot depend on a browser tab remaining open.

Performance targets require baseline tests, not guessed response-time commitments. Log real FTUE drop-offs, upload failures, skill selections, paid submissions, accepted shots, and incomplete projects only with appropriate privacy controls.

17. Suggested component tree and contracts for Codex

Presentation components: `CustomerShell`, `StudioWorkspace`, `ProjectRail`, `MediaCanvas`, `ShotSequence`, `DirectorPanel`, `DirectorComposer`, `AttachmentChips`, `AttachMenu`, `ReferencePicker`, `SkillPicker`, `VoiceControl`, `ShotContextBadge`, `ActivityDrawer`, `CostAuthorizationDialog`, `ReviewControls`, `MobileDirectorSheet`.

Business-facing services: `getAuthorizedProjectContext`, `listApprovedSkills`, `createSignedUpload`, `finalizeAsset`, `resolveAssetMentions`, `saveDraft`, `createDirectorMessage`, `prepareGenerationPlan`, `getPriceEstimate`, `authorizeCost`, `enqueueApprovedJob`, `getJobStatus`, `recordArtifactReview`. These are proposed interface names, not evidence of live endpoints.

Representative message shape (illustrative, not deployed):

```
{
  "workspace_id": "workspace-id",
  "project_id": "project-id",
  "stage": "production",
  "target": {"scene_id": "scene-id", "shot_id": "shot-id", "take_id": "take-id"},
  "body": "Make the emotion more restrained while preserving the camera.",
  "skill": {"id": "fix-shot", "version": "approved-version"},
  "references": [
    {"asset_id": "asset-id", "asset_version": 3, "role": "character_identity", "binding":
"required", "scope": "shot"}
  ],
  "draft_id": "draft-id",
  "client_request_id": "unique-client-request-id"
}
```

Safety separation: assistant messages and proposals are not job authorizations. A separate server-controlled authorization object binds an estimate version, operation scope, workspace/project, charge limit and idempotency identity. The UI does not contain provider credentials or privileged data access.

Suggested data additions only if not already present: `asset_reference_binding`, `composer_draft`, `message_skill_selection`, `approved_skill_version`, `voice_interaction_state`, `operation_proposal`, `cost_authorization`, and `ui_layout_preference`. Reuse existing normalized project/job records rather than creating duplicate sources of truth.

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18. Codex implementation plan: smallest safe patches

1. **Inspect known repo and current head.** Confirm which customer-facing routes and common composer already exist. Do not change the backoffice, payments, generation router or media worker just to deliver visual changes.
2. **Create reusable shell and three-pane layout.** Implement responsive split and collapse with existing component library where practical. Verify at desktop and phone sizes.
3. **Build composer in isolation.** Text drafting, local chips, `/` picker, `@` picker, `+` file selection; add accessibility and keyboard tests before backend calls.
4. **Connect authorized asset APIs.** Server-side checks, signed upload/finalize, role metadata, durable IDs and draft preservation. No direct provider upload from client if privileged tokens are required.

5. **Connect versioned skill registry.** Only approved skills; conditionally filter by task/stage. Skill selection remains non-billable.
6. **Attach to project context.** Selected scene, shot and take flow through the composer; switching scope triggers sensible draft retention and review.
7. **Connect voice in two stages.** First mic permission/transcription with editable transcript, then optional spoken replies. Realtime voice remains deferred until product and cost evidence warrants it.
8. **Connect paid-proposal gates last.** Use the existing authorized budget/job systems with explicit confirm, not a new direct call from Send.
9. **Verify changed surface only.** Snapshot production and briefing, run mobile interaction tests, verify no admin regressions; record which tests require real backend/provider service.

Change discipline: do not rewrite working route structures to match the demo. Prefer the smallest composable changes and preserve naming conventions, authorization policies, existing saved project data and approved design decisions.

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19. Acceptance criteria and test matrix

ID	Given / when	Expected result	Layer
C01	At homepage, choose Short Film	Brief starts without model selection	UI
C02	Enter brief, navigate away and return	Draft restored when authorized	Integration
C03	Attach a JPEG	Preview and filename shown	UI
C04	Assign Character Identity role	Binding stored with asset ID and scope	Integration
C05	Attach a script PDF	Document shown with script role	Integration
C06	Attach audio as Master Song	Timing anchor preserved in music-video route	Integration
C07	Upload fails midway	Draft survives; Retry offered	UI/Integration
C08	MIME mismatch	Client/server reject gracefully	Security
C09	Type `/` on a word boundary	Skill menu opens	UI
C10	Type `/sto`	Matching approved skill(s) filter	UI
C11	Arrow/Enter/Escape in menu	Focus and selection work predictably	Accessibility

ID	Given / when	Expected result	Layer
C12	Select `/fix-shot`	Skill chip inserted; no job launched	UI
C13	Remove skill chip	Message becomes ordinary text request	UI
C14	Type `@`	Only permitted project assets appear	Security/UI
C15	Mention asset	Stable asset/version ID included	Integration
C16	Switch selected shot	Visible context updates; unsent draft preserved safely	UI
C17	Ask to fix selected take	Proposal references exact version and constraints	Integration
C18	Cost estimate unavailable	Paid submission blocked	Billing
C19	Approve a repair	Only authorized affected job is queued	Integration
C20	Retry after unknown submission	Original request reconciled before resubmit	Reliability
C21	Approved neighboring shot exists	It survives unrelated correction	Domain
C22	Click View Activity	Status summary appears; hidden by default	UI
C23	Open mobile bottom sheet	Canvas and controls remain usable	Responsive
C24	Open mobile keyboard	Composer remains visible above keyboard	Device
C25	Mic permission denied	Type-instead option shown, no false listening state	Device
C26	Speech transcript returned	User can edit before sending consequential instruction	Voice
C27	Change orientation	Layout reflows, no clipped control	Responsive
C28	Private asset of other tenant referenced	Denied server-side	Security
C29	Duplicate Send event	No duplicate paid requests	Reliability
C30	Reviewer approves storyboard only	Motion approval remains outstanding	Domain

ID	Given / when	Expected result	Layer
C31	Full export fails after accepted takes	Retry finishing, preserve source media	Domain
C32	Reduced motion / keyboard nav	Visible focus, no essential motion-only instructions	Accessibility

QA evidence levels: screenshot review validates composition only; running prototype can validate interactions; production billing, access control, persistence and voice require integration/device tests. No user interviews or platform accessibility certification are claimed for this design.

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20. Open decisions, dependencies and non-goals

Decision	Current recommendation	Owner/verification before implementation
Initial storage	Reuse existing private asset storage after verifying active infrastructure	Engineering / security
Max upload size and media duration	Configure from measured storage/ provider constraints	Engineering / operations
Transcript provider/price	Cheapest verified route meeting target-language accuracy	Voice engineering
Continuous realtime voice	Defer; opt-in trial with strict session timeout and cost budget	Product / finance
Model defaults	Do not hardcode until benchmarked for each shot class	Model routing
Advanced node editor	Exclude from P0	Product
Desktop resizable panes	Implement if low-complexity and accessible; collapse-first fallback	Frontend
Mobile full timeline	Defer; emphasize review and corrections	Product
Record retention	Publish and enforce project/customer policy	Security / legal
Actual framework/repo baseline	Verify live repository before changes	Codex

No changes are authorized by this document alone. The companion film blueprint remains the source for backend responsibilities. The UI prototype is a visual and limited interaction demonstration, not production code.

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21. Visual handoff and files

The design package contains the screenshots shown here, the standalone HTML concept prototype and this specification. The HTML demonstrates panel structure, a searchable skill menu, project reference chips, a file-selection affordance, a View Activity toggle and a disabled cost-authorization example. File selection is **not** a persisted upload and microphone controls are **not** connected to actual transcription. Screenshots carry fictional project media and status labels for layout review.

Implementation check: Desktop Production Studio, Desktop Briefing, Mobile Composer, Mobile Skill Menu and Mobile Attachment Menu must be compared to the final code output. Maintain purpose and hierarchy, not every prototype pixel.

Definition of done: a customer can start/return to a project; attach and classify authorized references; type `/` and select an approved skill; use `@` for existing project assets; speak or type (once enabled); understand exact selected-shot scope; review an actionable proposal; and approve spending/creative output separately, all without exposing raw system thinking or accidentally replacing accepted work.

PLATE A | Desktop Production Studio: three-panel workspace

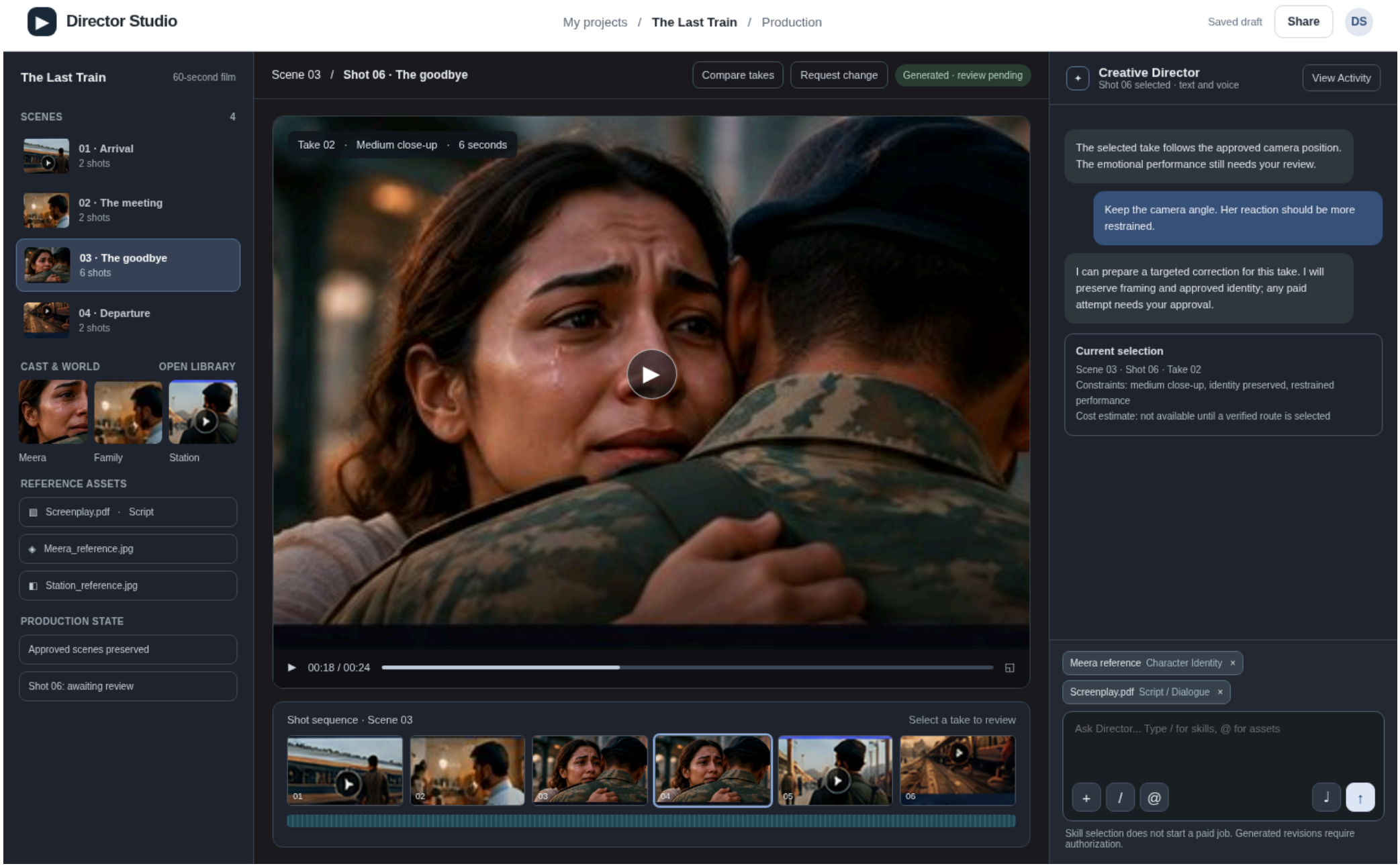


PLATE B | Desktop Voice Briefing: conversation and film brief

Director Studio

My projects / **The Last Train** / Creative briefing

Saved draft | Share | DS

CREATE A FILM

- **01 Creative brief**
- 02 Story development
- 03 Cast & world
- 04 Storyboard
- 05 Production
- 06 Edit & review

Each approved step is versioned. You can revisit earlier decisions without losing approved outputs.

Tell me about your film.

Speak naturally, type your idea, or attach a script, song or visual reference.

CREATIVE DIRECTOR

What would you like to make? Start with a sentence, or upload something you've already created.

A 60-second emotional story about an Indian soldier saying goodbye to his family.

CREATIVE DIRECTOR

Would you like the goodbye at home, at a railway station, or through a phone call?

A railway station. Realistic and understated.

CREATIVE DIRECTOR

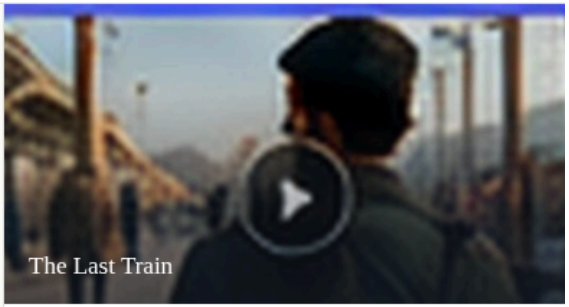
Understood. We can use your references to develop the characters and location without changing the approved story direction.

Meera reference Character Identity × | Screenplay.pdf Script / Dialogue ×

Add details... Type / for skills, @ to mention a reference

+ Attach | / Skills | @ References | ♩ Voice | Send

Uploads remain private. No video generation starts from a message alone.



Film brief · Draft

Format	**Short film**
Duration	**60 seconds**
Setting	**Railway station**
Emotional direction	**Realistic, restrained**
Next	**Review story treatment**

References

Character Identity · Meera_reference.jpg

Script / Dialogue · Screenplay.pdf

Reference roles define what should be preserved. The final film plan remains editable until approval.

Continue to story →

‹ The Last Train Shot 06 of 12 ···

Scene 03 · The goodbye Needs review



Review | Talk to Director | References

Keep the camera at eye level. The emotional performance should be restrained. Approved identity and wardrobe remain fixed.

Tell Director what to change Selected shot ↗

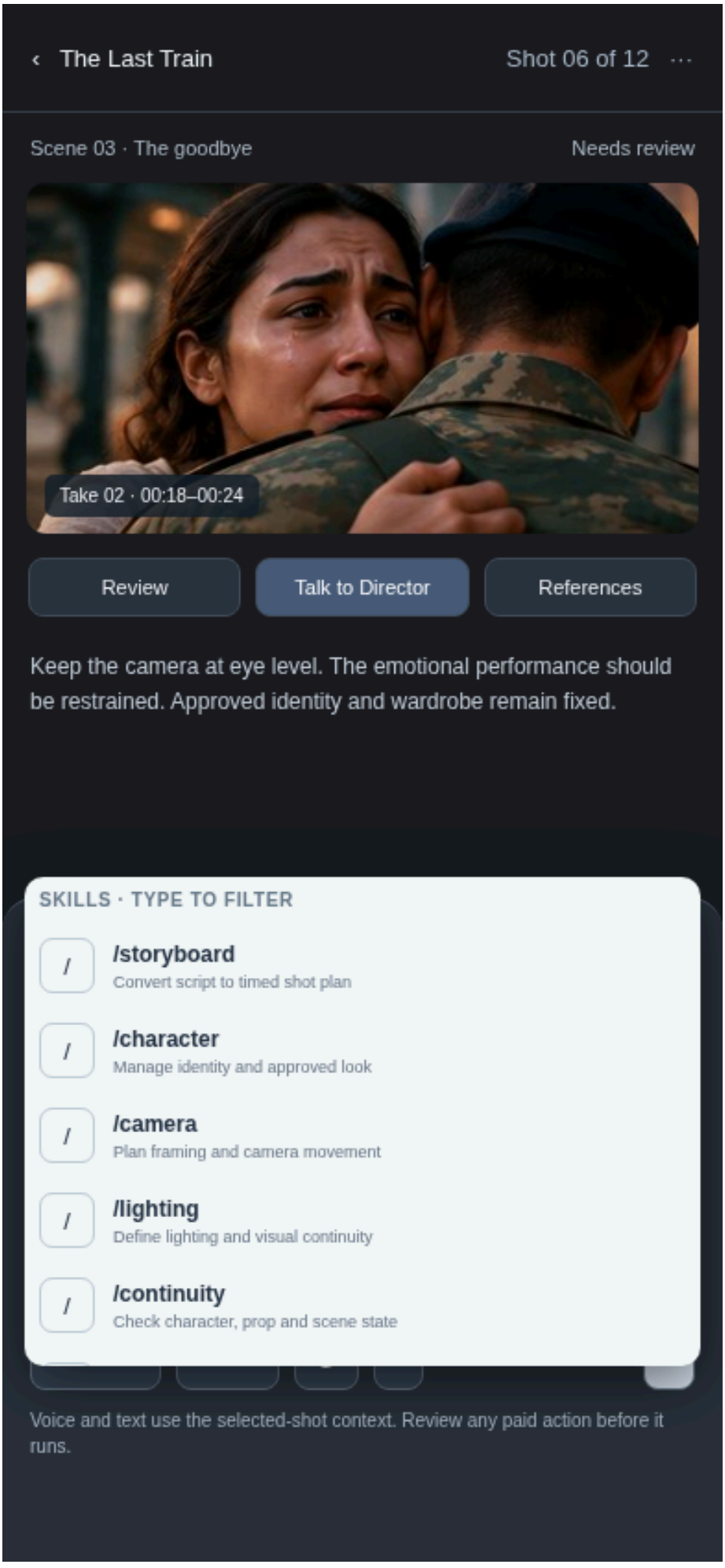
Meera reference Character Identity ×

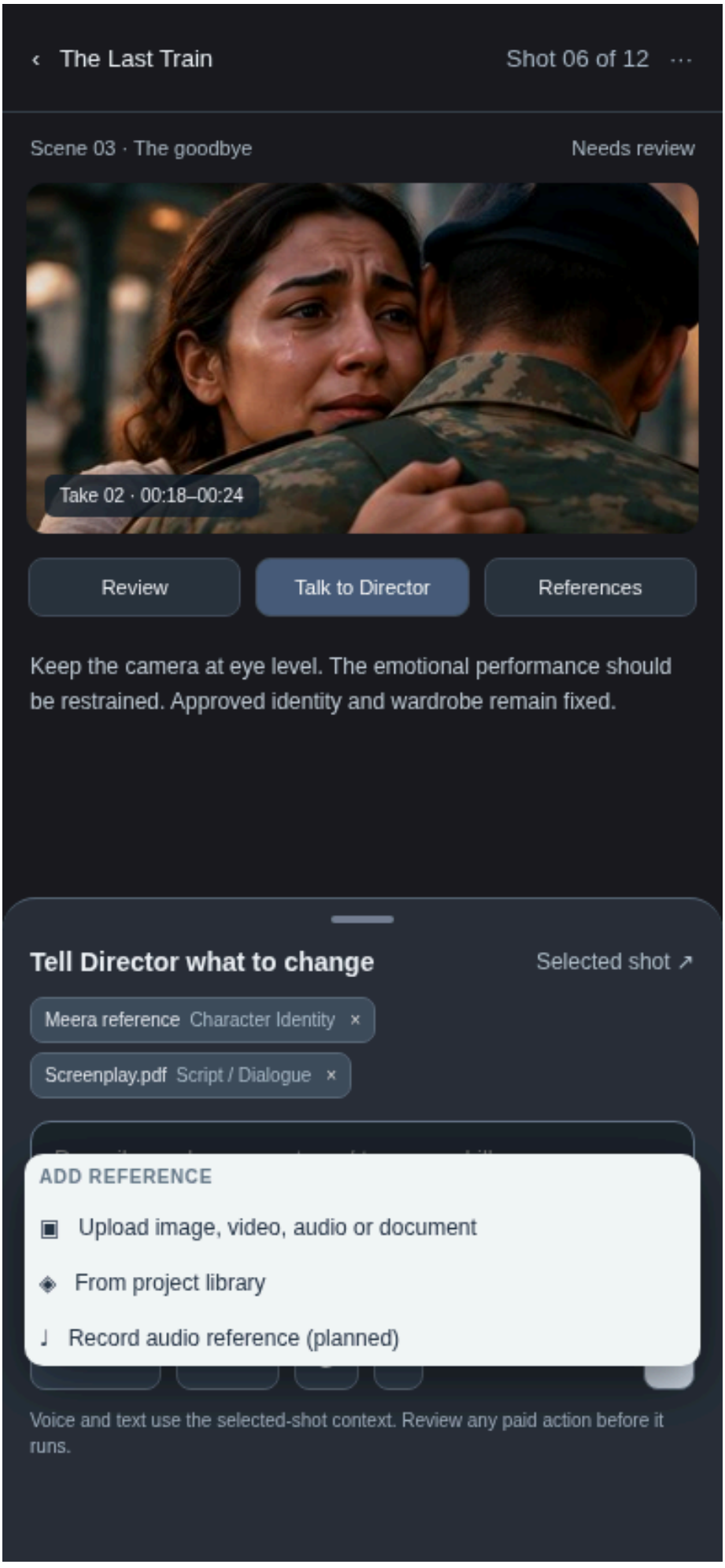
Screenplay.pdf Script / Dialogue ×

Describe a change, or type / to use a skill...

+ Attach / Skills @ ♩ ↑

Voice and text use the selected-shot context. Review any paid action before it runs.


‹ The Last Train
Shot 06 of 12 ···
Scene 03 · The goodbye
Needs review
Take 02 · 00:18–00:24
Review
Talk to Director
References
Keep the camera at eye level. The emotional performance should be restrained. Approved identity and wardrobe remain fixed.
SKILLS · TYPE TO FILTER
/storyboard
Convert script to timed shot plan
/character
Manage identity and approved look
/camera
Plan framing and camera movement
/lighting
Define lighting and visual continuity
/continuity
Check character, prop and scene state
Voice and text use the selected-shot context. Review any paid action before it runs.


‹ The Last Train
Shot 06 of 12 ···
Scene 03 · The goodbye
Needs review
Take 02 · 00:18–00:24
Review
Talk to Director
References
Keep the camera at eye level. The emotional performance should be restrained. Approved identity and wardrobe remain fixed.
Tell Director what to change
Selected shot ↗
Meera reference Character Identity ×
Screenplay.pdf Script / Dialogue ×
ADD REFERENCE
Upload image, video, audio or document
From project library
Record audio reference (planned)
Voice and text use the selected-shot context. Review any paid action before it runs.